



TRIOS CYBER

LABORATORY PRACTICE 19

Linux Wazuh Agent Deployment and Enrollment Verification

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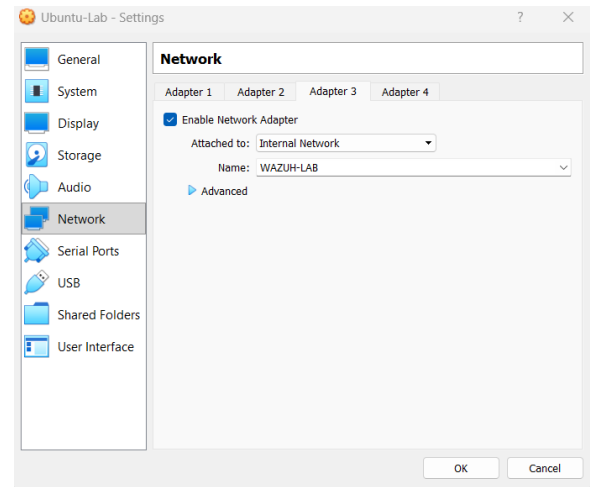
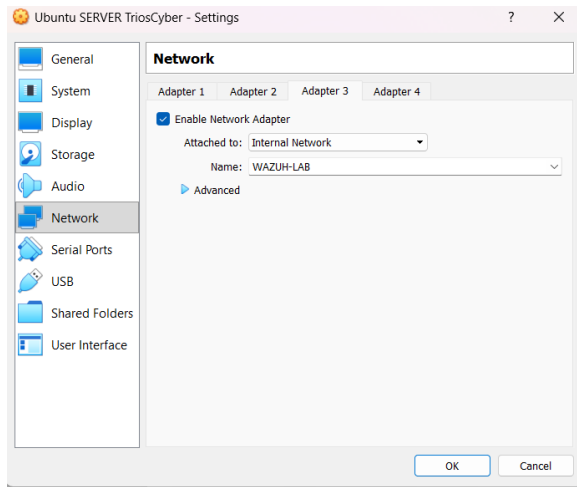
1. Laboratory Objective and Environment

The Day 19 activity deployed a Linux Wazuh agent on a separate Ubuntu virtual machine and enrolled it with the existing Wazuh server. The validation focused on isolated server-to-endpoint connectivity, agent installation and startup, and final confirmation that the endpoint was registered in Wazuh with an Active status.

Element	Observed value	Purpose / result
Wazuh server	10.10.20.10/24	Central manager and enrollment services
Ubuntu agent	10.10.20.20/24	Linux endpoint monitored by Wazuh
Internal network	WAZUH-LAB	Private VirtualBox communication path

2. VirtualBox Internal Network Configuration

An additional VirtualBox network adapter was enabled on both virtual machines and attached to the same Internal Network named WAZUH-LAB. This created a direct laboratory path between the Wazuh server and the Ubuntu endpoint while preserving the existing adapters used for other connectivity.



3. Dedicated Interface Addressing

The dedicated WAZUH-LAB interface was configured as enp0s9 on both systems. The Wazuh server used 10.10.20.10/24, while the Ubuntu endpoint used 10.10.20.20/24. The interface state was verified after the addresses were applied.

```
matthew@trioscyber:~$ sudo ip addr flush dev enp0s9
sudo ip addr add 10.10.20.10/24 dev enp0s9
sudo ip link set enp0s9 up
[sudo] password for matthew:
matthew@trioscyber:~$ ip -br a
lo                UNKNOWN    127.0.0.1/8 ::1/128
enp0s3            UP          10.10.10.10/24 fe80::6c39:68a2:ee11:51fe/64
enp0s8            DOWN        10.0.2.15/24 metric 100 fe80::a00:27ff:fe3d:82a6/64
enp0s9            UP          10.10.20.10/24 fe80::a00:27ff:feea:e392/64
```

```
matthew@matthew-VirtualBox: ~$ ip -br a
lo                UNKNOWN    127.0.0.1/8 ::1/128
enp0s3            UP          10.10.10.10/24 fe80::6c39:68a2:ee11:51fe/64
enp0s8            UP          10.0.3.15/24 fe80::6119:ea98:ffee:216f/64
enp0s9            UP          10.10.20.20/24 fe80::6119:ea98:ffee:216f/64
matthew@matthew-VirtualBox:~$ sudo ip addr flush dev enp0s9
[sudo] contraseña para matthew:
matthew@matthew-VirtualBox:~$ sudo ip addr add 10.10.20.20/24 dev enp0s9
matthew@matthew-VirtualBox:~$ sudo ip link set enp0s9 up
matthew@matthew-VirtualBox:~$ ip -br a
lo                UNKNOWN    127.0.0.1/8 ::1/128
enp0s3            UP          10.10.10.10/24 fe80::6c39:68a2:ee11:51fe/64
enp0s8            UP          10.0.3.15/24 fe80::6119:ea98:ffee:216f/64
enp0s9            UP          10.10.20.20/24 fe80::6119:ea98:ffee:216f/64
matthew@matthew-VirtualBox:~$
```

4. Bidirectional Connectivity and Wazuh Server Readiness

Connectivity was tested in both directions before enrollment. The Ubuntu endpoint reached the Wazuh server at 10.10.20.10 with four successful ICMP replies and 0% packet loss. On the server, the wazuh-manager service was restarted and returned active, and the server successfully reached the agent at 10.10.20.20 with 0% packet loss.

```
matthew@matthew-VirtualBox:~$ ip -br a
lo                UNKNOWN      127.0.0.1/8 ::1/128
enp0s3            UP            10.10.10.10/24 fe80::6c39:68a2:ee11:51fe/64
enp0s8            UP            10.0.3.15/24 fe80::6119:ea98:ffee:216f/64
enp0s9            UP            10.10.20.20/24
matthew@matthew-VirtualBox:~$ ping -c 4 10.10.20.10
PING 10.10.20.10 (10.10.20.10) 56(84) bytes of data.
64 bytes from 10.10.20.10: icmp_seq=1 ttl=64 time=1.84 ms
64 bytes from 10.10.20.10: icmp_seq=2 ttl=64 time=0.493 ms
64 bytes from 10.10.20.10: icmp_seq=3 ttl=64 time=0.341 ms
64 bytes from 10.10.20.10: icmp_seq=4 ttl=64 time=0.493 ms

--- 10.10.20.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3056ms
rtt min/avg/max/mdev = 0.341/0.792/1.842/0.609 ms
```

```
matthew@trioscyber:~$ sudo systemctl restart wazuh-manager
sleep 5
sudo systemctl is-active wazuh-manager

active
```

```
matthew@trioscyber:~$ ping -c 4 10.10.20.20
PING 10.10.20.20 (10.10.20.20) 56(84) bytes of data.
64 bytes from 10.10.20.20: icmp_seq=1 ttl=64 time=0.690 ms
64 bytes from 10.10.20.20: icmp_seq=2 ttl=64 time=0.614 ms
64 bytes from 10.10.20.20: icmp_seq=3 ttl=64 time=0.363 ms
64 bytes from 10.10.20.20: icmp_seq=4 ttl=64 time=0.300 ms

--- 10.10.20.20 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3102ms
rtt min/avg/max/mdev = 0.300/0.491/0.690/0.164 ms
```

The server was also checked for the Wazuh communication ports. TCP 1515 was listening through wazuh-authd for agent enrollment, while TCP 1514 was listening through wazuh-remoted for agent communication.

```
matthew@trioscyber:~$ sudo ss -ltnp | grep -E ':1514|:1515'
LISTEN 0      128          0.0.0.0:*        users:((("waz
uh-authd",pid=1707,fd=6))
LISTEN 0      128          0.0.0.0:*        users:((("waz
uh-remoted",pid=1811,fd=4))
```

5. Dashboard Access through the Existing SSH Tunnel

Because the Wazuh dashboard remained behind the VirtualBox NAT configuration, the existing SSH local-port-forwarding method was reused. Windows port 8443 was forwarded to 127.0.0.1:443 on the Wazuh server through the SSH connection on host port 2222.

```
PS C:\Users\User> ssh -N -L 8443:127.0.0.1:443 -p 2222 matthew@127.0.0.1
matthew@127.0.0.1's password:
```

Validation	Observed result	Meaning
Agent -> server	4/4 replies, 0% loss	10.10.20.20 reached 10.10.20.10
Wazuh manager	active	Server-side manager available
Agent ports	TCP 1514 and 1515	Communication and enrollment listeners
Dashboard path	Windows 8443 -> server 443	Browser access through SSH tunnel

6. Linux Agent Deployment from the Wazuh Dashboard

The Deploy new agent workflow was opened from the Wazuh Endpoints interface. Linux DEB amd64 was selected, the server address was set to 10.10.20.10, the agent name was defined as ubuntu-agent, and the default group was retained. Wazuh then generated the installation command for the Linux endpoint.

Select the package to download and install on your system:

LINUX

☐ RPM amd64 ☐ RPM aarch64

☒ DEB amd64 ☐ DEB aarch64

WINDOWS

☐ MSI 32/64 bits

macOS

☐ Intel ☐ Apple silicon

For additional systems and architectures, please check our [documentation](#).

Server address:

This is the address the agent uses to communicate with the server. Enter an IP address or a fully qualified domain name (FQDN).

Assign a server address

10.10.20.10

☒ Remember server address

Optional settings:

By default, the deployment uses the hostname as the agent name. Optionally, you can use a different agent name in the field below.

Assign an agent name

ubuntu-agent

The agent name must be unique. It can't be changed once the agent has been enrolled.

Select one or more existing groups

Default

Run the following commands to download and install the agent:

```
wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.7-1_amd64.deb && sudo WAZUH_MANAGER='10.10.20.10' WAZUH_AGENT_NAME='ubuntu-agent' dpkg -i ./wazuh-agent_4.14.7-1_amd64.deb
```

Requirements

- You will need administrator privileges to perform this installation.
- Shell Bash is required.

Keep in mind you need to run this command in a Shell Bash terminal.

7. Wazuh Agent Package Installation

The generated command was executed on the Ubuntu endpoint. The Wazuh agent package version 4.14.7-1 for amd64 was downloaded from packages.wazuh.com and installed with the manager address and agent name supplied by the dashboard workflow.

```
matthew@matthew-VirtualBox:~$ wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.7-1_amd64.deb && sudo WAZUH_MANAGER='10.10.20.10' WAZUH_AGENT_NAME='ubuntu-agent' dpkg -i ./wazuh-agent_4.14.7-1_amd64.deb
--2026-09-21 00:13:48-- https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.7-1_amd64.deb
Resolving packages.wazuh.com (packages.wazuh.com)... 54.230.144.106, 54.230.144.26, 54.230.144.88, ...
Conectando con packages.wazuh.com (packages.wazuh.com)[54.230.144.106]:443... conectado.
Petición HTTP enviada, esperando respuesta... 200 OK
Longitud: 13222832 (13M) [application/vnd.debian.binary-package]
Guardando como: "wazuh-agent_4.14.7-1_amd64.deb.1"

wazuh-agent_4.14.7- 100%[=====] 12,61M 30,0MB/s en 0,4s

2026-09-21 00:13:48 (30,0 MB/s) - "wazuh-agent_4.14.7-1_amd64.deb.1" guardado [13222832/13222832]

[sudo] contraseña para matthew:
(Leyendo la base de datos ... 226146 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../wazuh-agent_4.14.7-1_amd64.deb ...
Desempaquetando wazuh-agent (4.14.7-1) sobre (4.14.7-1) ...
Configurando wazuh-agent (4.14.7-1) ...
```

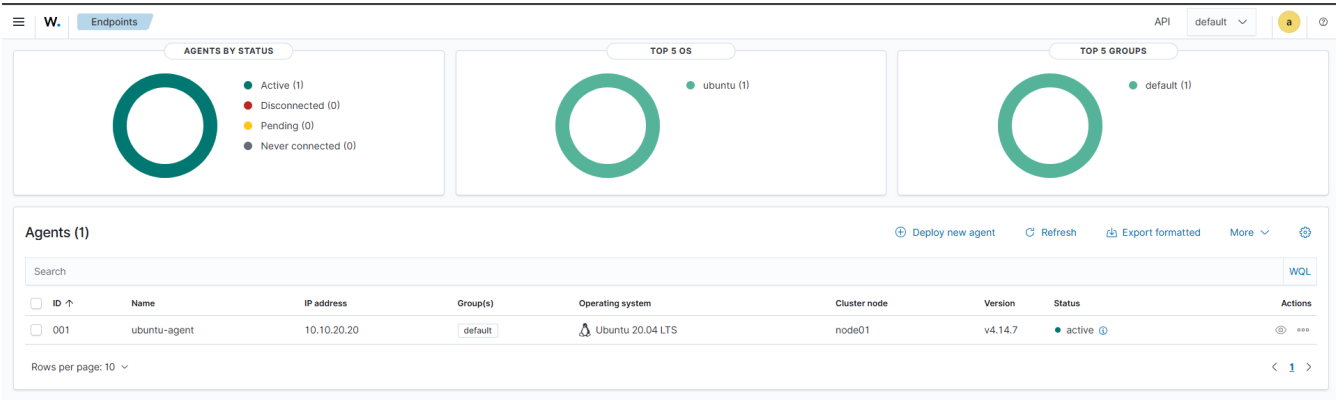
8. Agent Service Initialization

After package installation, systemd was reloaded and the wazuh-agent service was enabled and started. These commands prepared the endpoint to initiate communication with the Wazuh manager using the enrollment configuration created during deployment.

```
Processando dependências para systemd (245.4 / ubuntu20.04) ...
matthew@matthew-VirtualBox:~$ sudo systemctl daemon-reload
matthew@matthew-VirtualBox:~$ sudo systemctl enable wazuh-agent
matthew@matthew-VirtualBox:~$ sudo systemctl start wazuh-agent
```

9. Final Agent Registration and Active Status

The Wazuh Endpoints view was refreshed after enrollment. One agent was displayed and its status was Active. The registered endpoint appeared as ID 001 with the name ubuntu-agent, IP address 10.10.20.20, group default, Ubuntu 20.04 LTS, cluster node node01, and Wazuh agent version v4.14.7.



Element	Observed value	Purpose / result
Agent ID / Name	001 / ubuntu-agent	Registered endpoint
IP / OS	10.10.20.20 / Ubuntu 20.04 LTS	Expected laboratory endpoint
Version / Group	v4.14.7 / default	Installed agent and assigned group
Final status	Active	Successful manager-agent communication

10. Conclusions

The Linux Wazuh agent deployment was completed successfully. The two virtual machines communicated through the WAZUH-LAB internal network, the Wazuh server was available on the required agent ports, and the Ubuntu package was installed and started using the deployment command generated by the dashboard.

The final Endpoints view confirmed that ubuntu-agent was enrolled and reporting as Active at 10.10.20.20. This result satisfies the Day 19 requirement to deploy a Linux Wazuh agent and verify that the endpoint becomes active in agent management.